

A Manifesto for Renewing the Practice of Mathematics

This text follows from some reflections I have had on the current impact of AI on mathematics and on the reactions I have seen. It is also a follow-up to my first essay.

It is based on one premise: there exists a *credit* system that forms the foundation of the mathematical academic system. By credit system, I mean the following: when someone writes a new mathematical article that person gets *credit* for it. The amount of *credit* is generally understood to increase with how difficult the task is believed to have been or how original the ideas. *Believed* in the sense that this is not measured by some intrinsic metric, but by the perception of a sufficiently influential group of people belonging to the academic system.

People with enough *credit* gain various social or economic advantages: jobs, prizes, grants, invitations, admiration, and so on. Some of these advantages, in turn, make the accumulation of *credit* somewhat self-sustaining.

I believe that this system is antithetical with the current capabilities of AI, as long as we maintain a minimum degree of intellectual honesty and acknowledge that people should have the right either to use AI or not to use it at all in their work. I believe that this system poses many problems and creates the wrong incentives.

Imagine a mathematician or a group of mathematicians working on some original problem without AI, as they should have every right to do. They are now strongly incentivized to keep this project as secret as possible, because they are highly exposed to someone else solving it extremely quickly using AI. This situation, of course, already existed before, but the cost of getting into a problem, finding paths toward solutions, and writing the article was so high that this type of situation ended up being relatively rare. These costs are now extremely low, and I believe they will continue to decrease.

Imagine someone with enough credit to have access to a large amount of grant money. This person then has the power to leverage a large amount of AI compute to solve problems that are considered extremely difficult, with the ultimate goal of getting more credit for it. Once again, to put it cynically, this already existed before through the employment of non-permanent researchers, but once again, the scale is completely different.

Imagine someone discovering a proof of a problem considered very difficult in a completely automated way, then digesting and rewriting the whole thing, and getting credit for it without any AI disclosure. It is easy to do now. I believe it already happens. I never count on other people's honesty, particularly when there are thousands of people incentivized based on their performance.

Today, a colleague of mine received by e-mail a proof of one of his conjectures from someone outside the academic system, with no previous record of working on the subject. My colleague is not naive: he believes the proof is AI-generated. Still, the person who generated it and sent it had every right to do so. The “mistake” of my colleague was, in a sense, simply to have written down their conjecture in a previous article. In a credit-based academic system, we will then be incentivized to hide our conjectures, our problems, and our ideas. Is that a healthy way forward?

Think for a moment about what happens when someone posts on arXiv, before you do, an article solving the problem you were working on. You are disappointed, but why? The experience, the understanding, the broader perspective you gained from working on it did not disappear. These were not stolen from you. No, you mourn the *credit* you will not get. Your ego will be unsatisfied because you will not be the one who “solved it”, your career might not get the boost you expected. But what if this did not count anymore? What if this did not matter? Wouldn't it be a relief?

In addition, if someone produces a proof of a good result without acknowledging AI use, by what criteria will you decide that this person did not “cheat”? Their previous record? Then there is no room for improvement. And what about the PhD? Should people need the stamp “able to do great mathematics” in order to have the privilege of being believed if they produce a breakthrough? What a nice world that would be !

I have read many suggestions on what to do. A lot of them are nothing but mitigation to save a system which is now obsolete.

Some people respond with a certain degree of gatekeeping, but, as I have already said, no individual or group of individuals has any ownership rights over mathematics. If companies, or individuals outside academia, want to solve mathematical problems by whatever means they choose, they have every right to do so. They even have the right not to follow academic conventions.

I have also seen people talk about the alignment of the mathematical community with AI or with AI companies. But do they forget that this “mathematical community” is a very abstract and loosely organized concept, bringing together tens of thousands of people with very different ideologies, incentives, and cultures? Why should anyone believe they can speak on behalf of such a large group of people? To me, this reflects a deep form of ingrained elitism.

I don't believe that social pressure should be used to enforce some unwritten rules, decreed by people whose authority comes from the *credit* they have accumulated, especially when those rules serve to preserve a system from which they themselves greatly benefit. If I have learned one thing recently, it is that being extremely good at mathematics is no guarantee of common sense nor ground intelligence.

My belief is that we should give up the concept of *credit*. I also believe that this is where we are heading, whether we want it or not.

So shall we do ?

From now on, I will give up any credit I could get (I was only getting a little anyway, so not much of a loss for me XD). My future projects will take place out in the open, fully shared on GitHub from the very beginning to the end. Everyone will be welcome to participate, in any way they want, but none of the participants will receive any credit for their participation. At least, not in the sense rewarded by the academic system.

The goal will be to produce a digested account of a mathematical topic, somewhat like a book, but including original results credited to no one. Right now, people are flooding arXiv and journals with AI-aided or AI-generated articles because they want to get the credit for it. I don't want to add to the noise. I don't want credit for anything. I want to build something that makes sense.

From now on, I will always assume AI use, even when it is not disclosed. For a short period of time, if someone explicitly states that no AI was used, I will give them the benefit of the doubt. But in my ideal, uncredentialed world, this question should no longer matter.

As I already stated in my last essay, I will finish my current projects as usual. After that, I believe I will never submit an article to a journal again, because this whole system no longer makes sense to me. I am fine staying an assistant professor. I am already far luckier than the majority of my fellow citizens.

I, of course, expect that the majority will not agree with me, but I hope that at least some people will join.